







International Conference on Wireless Communication for Military Applications WiCOMM-2026

Fail-Closed or Fail-Degraded? A Simulation Study of Forced Assurance Loss in Contested Multi-Hop QKD Networks

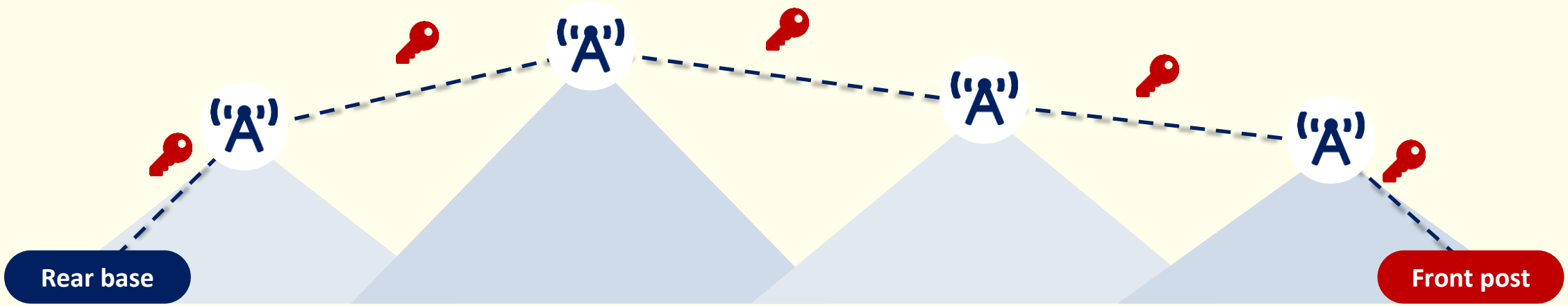
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Motivation: When Quantum Key Runs Out

Out of key, a network must refuse service or quietly weaken it



Every relay re-encrypts the key with a one-time pad, so an *n*-hop path pays for key on **every hop**, and a free-space hop makes only about 10^2 bit/s.

Fail closed

Refuse the request, as a layered PQC design does.
The commander sees an outage and can act.

Fail degraded

Fall back to a weaker cipher, as ITU-T Y.3803 allows.
Nobody notices until the audit.

Research Gap and Contributions

Prior work counts service, not the ITS protection it lost

Line of prior work	Served?	ITS loss?
Routing which path a request takes	✓	✗
Quality of service who gets key first	✓	✗
Exhaustion and fallback what happens when key runs out	✓	✗
Adversary models depletion, attack radius	✓	✗

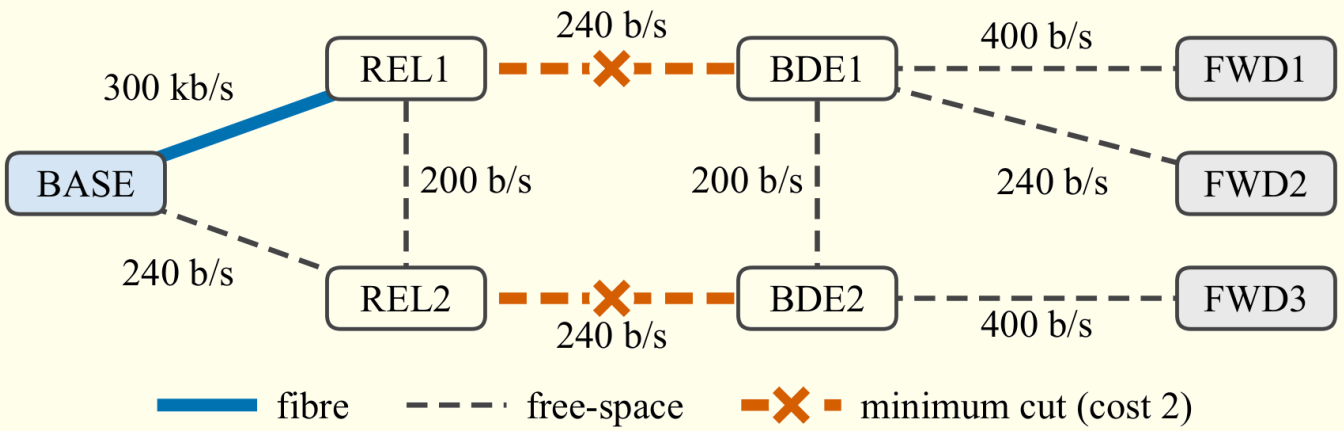
The missing column. No study we found reports how much traffic entitled to ITS key is served below ITS on a multi-hop network under attack.

What this paper adds

- **A threat-aware model** where stored key outlives generation
- **Forced assurance loss F** and the downgrade-forcing cost k*
- **A cut condition** that bounds how long compliant service lasts
- **Evidence on five graphs** that forcing needs capability, a cut and time
- **An audit** of a learned controller, and accounting checked against published measurements

System Model: Formation, Key Grades and Adversary

Every session to the front crosses one of two 240 bit/s links



Paper Fig. 1(a). Orange crosses mark the two-link minimum cut.

Three failure surfaces, kept separate

Quantum channel
QBER past 0.075 stops fresh key

Classical transport
no relaying, no fallback

Key service
no KEM fallback setup

The assurance ladder

G_1	One-time pad	ITS
G_2	QKD-keyed cipher	ITS key
G_3	QKD ⊕ ML-KEM	below ITS
G_4	ML-KEM only	post-quantum

Floors C2 → G_2 ISR → G_1 LOG → G_3

The attacker raises the QBER on links it buys until they abort.
Stored key survives, so it must outlast the stock.

Metrics and Experimental Setup

Availability hides the damage, so we measure the grade served

Fail-degraded, two-link cut held for 24 h, 2% fill

6.6% at grade

2.5% not served



97.5% counted as available

A = Q + L

Availability splits into service at its authorised grade (Q) and below it (L).

F forced assurance loss

Requests entitled to ITS key that are served below ITS. Blocked ones stay in the count.

$T^* = \Sigma U_e(0) / D$

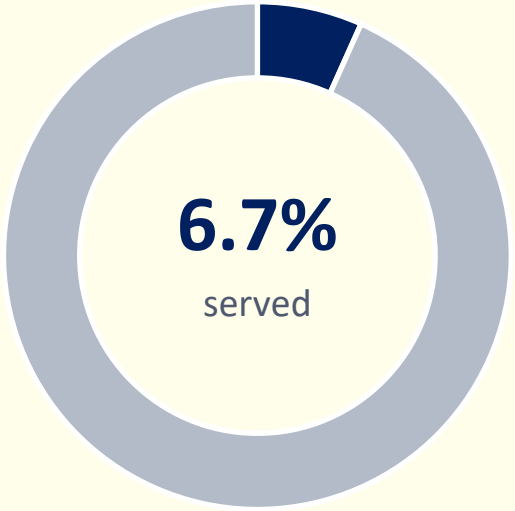
The longest compliant service can last under a held cut: 1.44 h at 2% fill.

Tested with 24 h attacks, 12 paired seeds and 95% bootstrap intervals, behind a 43-check gate and 127 regression tests.

Key Result: The Fail-Mode Exchange Rate

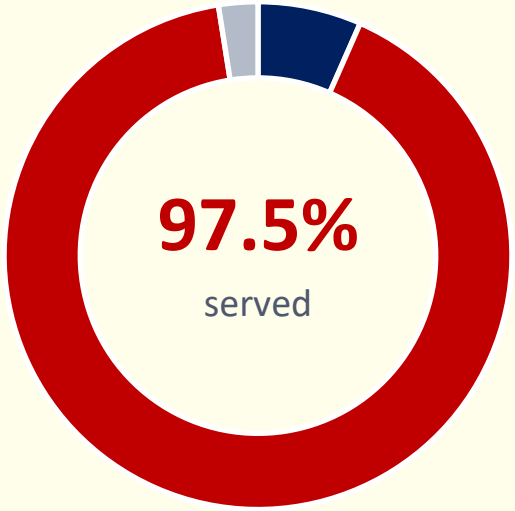
Same attack: refuse nearly everything, or serve it below grade

Fail closed



Refuses the rest: a visible outage

Fail degraded



Only 6.6% at its grade

at authorised grade below floor (all classes) refused

90.1%

of requests entitled to ITS key are served without it (95% CI 89.2–91.0, 12 seeds)

93.2%

of served sessions run on ML-KEM-768 alone

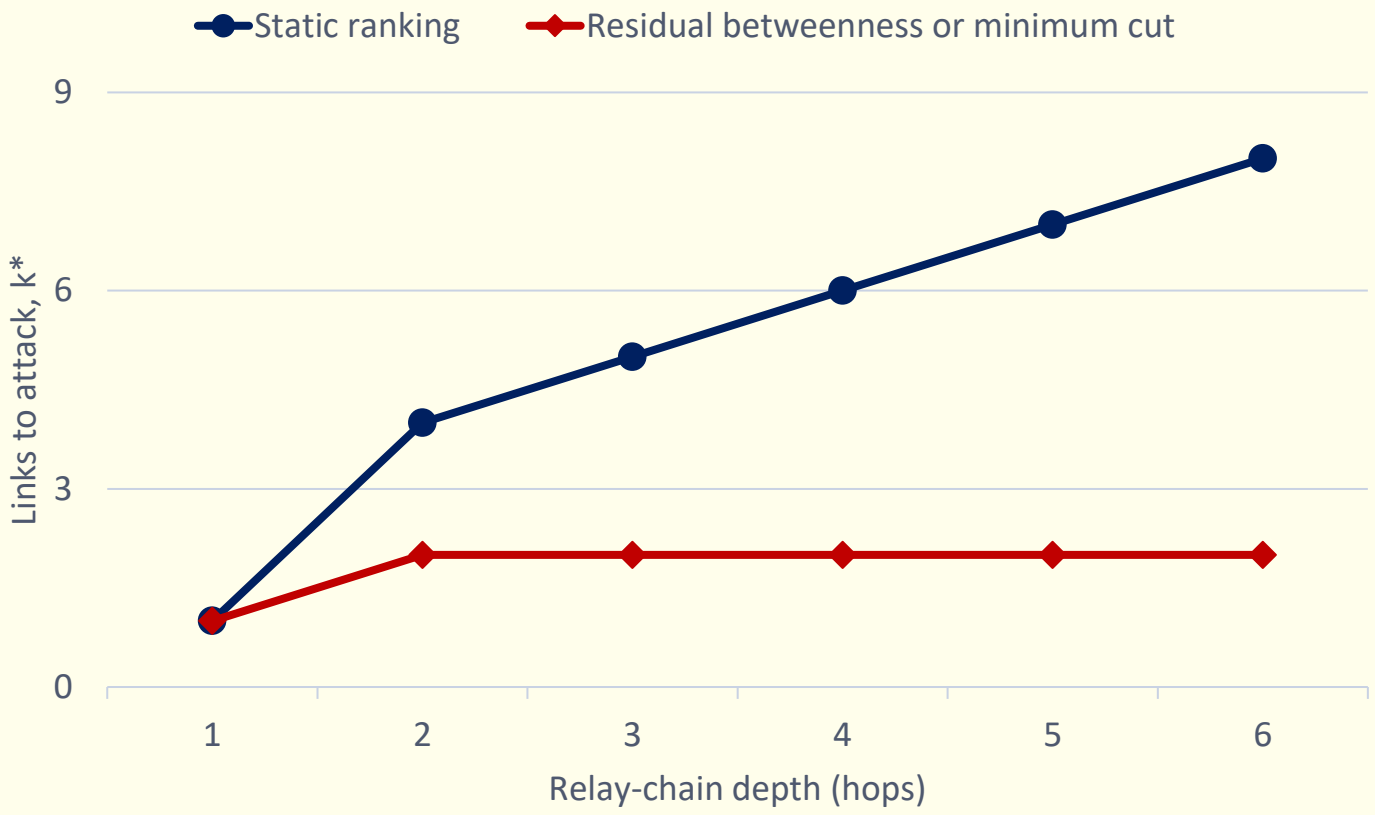
87.6–92.9%

forced loss on all five graphs when the cut is held

What Forcing Requires: Capability, Cut and Endurance

Forcing needs capability, a cut and time. Remove one, it fails

k*: links an attacker must buy to force F ≥ 50% in 24 h (fail-degraded, 2% fill, 8 seeds)



Capability

A link aborts only once **Δe ≥ 0.045**; below that (γ = 1), under 0.1% is forced.

A cut

A smart attacker needs only the **2-link cut**, however deep the chain.

Time

First forcing after **about 1.5 h** at 2% fill; from full stores, nothing in 24 h.

Routing adds 0.12 points of Q; floors cut F to 63.6% by denying logistics key; a learned controller gains little and raises payload exposure.

Conclusions and Recommendations for Operators

Report the grade of what you serve, not only that you served it

Three rules for operators

- **Report Q and F** next to availability A.
- **Keep stock behind the minimum cut** for T* hours of the attacks you fear.
- **Set fail modes per traffic class**, and put a price on every exception.

Open and reproducible. Every number comes from the released simulator, configurations and per-seed results.

Thank you

Questions are welcome

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github.com/LakshyaJ1/Quantum_ResearchPaper